

Chapter 1

Linear Notions

Theorem 1 (Infinite interior points). *Let $A, C \in \mathbf{P}$ with $A \neq C$. Then there exist infinitely many points strictly between A and C .*

Proof. wasd □

Theorem 2 (Infinite outward extension). *Let $A, B \in \mathbf{P}$ with $A \neq B$. Then there exist infinitely many points $C_1, C_2, C_3, \dots$ such that*

$$\mathcal{B}(A, B, C_1), \quad \mathcal{B}(A, C_1, C_2), \quad \mathcal{B}(A, C_2, C_3), \quad \dots$$

Proof. wasd □

Together, these two lemmas show that from any segment AB ($A \neq B$) we can generate infinitely many points in both directions along the line determined by A and B . Thus the plane $\mathcal{P}$ is necessarily infinite.

1.1 Defintion of Linear Notions

A segment is a subset of $\mathbf{P}$ defined as the union of two discint points and all points between them. Symbolically, that is

Definition 1 (Segment).

$$\forall A, B \in \mathbf{P}, A \neq B \implies AB := \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}.$$

A ray pointing from point A to a discint point B is a subset of $\mathbf{P}$ denoted as $\overrightarrow{AB}$. It is defined as the union of A, B , all points between A and B , and all points such that B is between A and that point, that is

Definition 2 (Ray).

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overrightarrow{AB} := \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The line $\overleftrightarrow{AB}$ is the subset of $\mathbf{P}$ consisting of the points A, B , all points between A and B , all points where A is between it and B , and all points where B is between it and A . That is,

Definition 3 (Line).

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overleftrightarrow{AB} := \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

1.2 Properties of Linear Notions

Theorem 3 (Distinctness of Betweenness). *If $\mathcal{B}(A, B, X)$ holds, then A , B , and X are three distinct points:*

$$\forall A, B, X \in \mathbf{P}, \mathcal{B}(A, B, X) \implies (A \neq B \wedge B \neq X \wedge A \neq X).$$

Proof. We proceed by contradiction. Suppose $\mathcal{B}(A, B, X)$ holds.

First, assume $A = X$. Substituting A for X in our hypothesis, we obtain $\mathcal{B}(A, B, A)$, which implies $A = B$. If $A = B$, then the relation $\mathcal{B}(A, B, X)$ is equivalent to $\mathcal{B}(B, A, X)$. However, by the Order of Betweenness, $\mathcal{B}(A, B, X) \implies \neg \mathcal{B}(B, A, X)$. Here, we reach a contradiction and conclude our hypothesis must be false and thus $A \neq X$.

Next, we test the assumption $A = B$. Under this assumption, we substitute our hypothesis to obtain $\mathcal{B}(B, A, X)$. Similarly, the Order of Betweenness shows these cannot be simultaneously true, and thus our assumption must be false, meaning $A \neq B$.

Finally, by the Transitivity of Equality, $B \neq X$. Thus proving the result. $\square$

Theorem 4 (Segment Symmetry). *For any two points A, B , the segment AB is equal to the segment BA .*

$$\forall A, B \in \mathbf{P}, AB = BA.$$

Proof. By the definition of a line segment, we have:

$$AB = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}$$

and

$$BA = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}.$$

First, by the commutative property of sets, the endpoints are identical: $\{A, B\} = \{B, A\}$.

Second, we examine the sets of interior points. By the **Symmetry of Betweenness**, the relation $\mathcal{B}(A, X, B)$ holds if and only if $\mathcal{B}(B, X, A)$ holds. Formally, this implies:

$$\{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}.$$

Since AB and BA are defined as the union of identical sets, they contain the same points. By the principle of extensionality in set theory, $AB = BA$. $\square$

Theorem 5 (Line Symmetry). *For any two distinct points A and B , the line $\overleftrightarrow{AB}$ is equal to the segment $\overleftrightarrow{BA}$:*

$$\forall A, B \in \mathbf{P}, \overleftrightarrow{AB} = \overleftrightarrow{BA}.$$

Proof. By the definition of a line, we have:

$$\overleftrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}$$

and

$$\overleftrightarrow{BA} = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, B, A) \vee \mathcal{B}(B, X, A) \vee \mathcal{B}(B, A, X)\}.$$

First, by the commutative property of sets, the endpoints are identical: $\{A, B\} = \{B, A\}$.

Second, we demonstrate that the three cases of betweenness in $\overleftrightarrow{AB}$ are logically equivalent to those of $\overleftrightarrow{BA}$ using the **Symmetry of Betweenness**:

- $\{X \in \mathbf{P} \mid \mathcal{B}(X, A, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, A, X)\}$
- $\{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}$
- $\{X \in \mathbf{P} \mid \mathcal{B}(A, B, X)\} = \{X \in \mathbf{P} \mid \mathcal{B}(X, B, A)\}$

Since each condition for membership in $\overleftrightarrow{AB}$ is logically equivalent to a condition for membership in $\overleftrightarrow{BA}$, the sets contain the same points. Thus, by the principle of extensionality, $\overleftrightarrow{AB} = \overleftrightarrow{BA}$. $\square$

Theorem 6 (Ray Asymmetry). *For any two distinct points $A, B \in \mathbf{P}$, the ray $\overrightarrow{AB}$ is not equal to the ray $\overrightarrow{BA}$:*

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overrightarrow{AB} \neq \overrightarrow{BA}.$$

Proof. We proceed by contradiction. Assume that the rays for some distinct points $A, B \in \mathbf{P}$, $\overrightarrow{AB} = \overrightarrow{BA}$.

By the definition of a ray, we have:

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\},$$

and

$$\overrightarrow{BA} = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A) \vee \mathcal{B}(B, A, X)\}.$$

By the **Outer Point Axiom**, there must exist a point Y such that $\mathcal{B}(A, B, Y)$. By the definition of $\overrightarrow{AB}$, it follows that $Y \in \overrightarrow{AB}$. Under our initial assumption that the rays, Y must be an element of $\overrightarrow{BA}$.

By the Distinctness of Betweenness, the relation $\mathcal{B}(A, B, Y)$ implies that $A \neq Y$ and $B \neq Y$. Therefore, $Y \notin \{A, B\}$.

Since $Y \in \overrightarrow{BA}$ but is not an endpoint, it must satisfy $\mathcal{B}(B, Y, A)$ or $\mathcal{B}(B, A, Y)$.

However, because we know that $\mathcal{B}(A, B, Y)$ is true, by Order of Betweenness we have $\mathcal{B}(A, B, Y) \implies \neg \mathcal{B}(B, A, Y)$, and $\mathcal{B}(A, B, Y) \implies \neg \mathcal{B}(A, Y, B)$, which by the Symmetry of Betweenness is logically equivalent to $\mathcal{B}(A, B, Y) \implies \neg \mathcal{B}(B, Y, A)$.

Thus, both conditions for Y to be an element of $\overrightarrow{BA}$ are false. Therefore, the rays $\overrightarrow{AB}$ and $\overrightarrow{BA}$ are not equal, thus proving the desired result. $\square$

Theorem 7 (Union of Opposite Rays). *For any two points $A, B \in \mathbf{P}$,*

$$\overleftarrow{AB} \cup \overrightarrow{AB} = \overleftrightarrow{AB}.$$

Proof. By the definition of a ray, we have:

$$\overleftarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B)\},$$

and

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The union of $\overleftarrow{AB}$ and $\overrightarrow{AB}$ therefore consists of the set, $\{B, A\}$, and all points X such that $\mathcal{B}(X, A, B)$ or $\mathcal{B}(A, X, B)$, or $\mathcal{B}(A, B, X)$.

Note that this is the same as the line $\overleftrightarrow{AB}$, as it is defined as the set

$$\overleftrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

This set is equal to the union described above. Thus, the union of the rays $\overleftarrow{AB}$ and $\overrightarrow{AB}$ is equal to the line $\overleftrightarrow{AB}$. $\square$

Theorem 8 (Intersection of Opposite Rays). *For any two points $A, B \in \mathbf{P}$,*

$$\overleftarrow{AB} \cap \overrightarrow{AB} = AB.$$

Proof. By the definition of a ray, we have:

$$\overleftarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B)\},$$

and

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The intersection of $\overleftarrow{AB}$ and $\overrightarrow{AB}$ therefore consists of the set, $\{B, A\}$ and all points X between A and B , that is $\mathcal{B}(A, X, B)$.

This intersection consists of the same points contained in the segment AB , as

$$AB = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}.$$

Thus, the union of the rays $\overleftarrow{AB}$ and $\overrightarrow{AB}$ is equal to the segment AB . □

1.2.1 Midpoints and Bisectors

The *midpoint* of a segment AB is the point M such that $AM \cong MB$ and $[AMB]$.

Axiom 1 (Midpoint Existence). $\forall A, B \in \mathbf{P}, A \neq B \implies \exists M \in \mathbf{P}: (\mathcal{B}(A, M, B)) \wedge (AM \cong MB)$.

Axiom 2 (Midpoint Uniqueness). $\forall A, B \in \mathbf{P}, A \neq B \implies \forall M, N \in \mathbf{P}: (AM \cong MB) \wedge (\mathcal{B}(A, M, B)) \wedge (AN \cong NB) \wedge (\mathcal{B}(A, N, B)) \implies M = N$.

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By removing betweenness as a criteria, we obtain the set of points $\{X \mid AM \cong MB\}$. This set is a line that is perpendicular to the segment AB . Alternatively, we can define the perpendicular bisector as the line containing the midpoint M of segment AB

1.3 Parallelism

Two lines are parallel if they have no point in common. In our set-theoretic foundations, this is to mean that the intersection of the lines is the empty set. Because we define parallelism as a binary relation on lines, we first must have a set of all lines to work with. Thus, we let $\mathbf{P}$ be the domain called the set of all points and define $\mathbf{L}$ to be the collection of all subsets of $\mathbf{P}$ that consist of the line $\overleftrightarrow{AB}$ for a any given two points $A, B \in \mathbf{P}$. Formally, that is:

$$\mathbf{L} := \{l \subseteq \mathbf{P} \mid \exists A, B \in \mathbf{P} (A \neq B \wedge l = \overleftrightarrow{AB})\}.$$

Definition 4 (Parallel). *Given any lines $l, m \in \mathbf{L}$, we say l is parallel to m and write $l \parallel m$ if and only if they are the same set of points or their set intersection is empty.*

$$\forall l, m \in \mathbf{L}, l \parallel m \iff (l = m) \vee (l \cap m = \emptyset).$$

add more here